

11 Quick diagnostic matrix

Use this page as a first-pass field conversation guide. Confirm with measurements before acting.

Finding	Possible process	Next checks
Tip burn strongest in dry or low-flow areas	Water deficit and chloride concentration	Emitter output pressure soil moisture water Cl and leaf Cl
Tip burn strongest in wet low areas	Poor drainage weak roots salt accumulation or disease	Root condition soil oxygen or drainage ECe Cl Na and Phytophthora history
Low leaf K with low soil K	Likely supply limitation especially in low-CEC soil	Sampling quality CEC texture roots water and crop load
Low leaf K with adequate soil K	Restricted uptake or demand effect	Moisture salinity roots drainage crop load Mg Ca Na and sampling timing
High water SAR and slow infiltration	Sodium-related dispersion risk	Water EC Ca Mg Na soil SAR structure and site-specific amendment need
High ECe but moderate leaf Cl	General salinity may dominate or exposure is recent	Full ion panel moisture trend root growth and seasonal leaf follow-up
High leaf Cl but modest bulk-soil Cl	Sampling mismatch localized salt or historical accumulation	Sample wetted root zone by depth confirm leaf cohort water source and wetting pattern
Chlorosis on new leaves in high-pH wet soil	Micronutrient unavailability plus weak roots	pH bicarbonate drainage roots Fe Zn Mn and irrigation timing
Organic amendment followed by burn	Salt ammonium immaturity or excessive local rate	Amendment EC Na Cl ammonium maturity rate placement and soil ECe
Biological product with no field response	Poor habitat strain mismatch low viability or wrong target	Root-zone constraints product identity storage untreated comparison and measured endpoint

Core field data bundle for future DST tools

- Standardized leaf analysis with date leaf age cultivar rootstock block and crop load.
- Irrigation water EC pH bicarbonate chloride sodium calcium magnesium and SAR by source and season.
- Soil texture organic matter pH ECe chloride SAR or exchangeable sodium CEC and nutrients sampled in the active wetted root zone and by depth where useful.
- Irrigation system type run time frequency emitter output pressure uniformity wetted area and recent leaching or rainfall.
- Root and soil observations including compaction infiltration drainage aeration root color root density mulch depth and disease history.